

EMPIRICAL RULE :-

The empirical rule is a shortcut for normal distributions that says how much data lies within 1, 2 and 3 standard deviations of the mean : about 68%, 95% and 99.7% respectively.

* WHAT EMPIRICAL RULE SAYS?

FOR a variable X that is (approximately) normally, distributed with mean μ and standard deviation σ

- BETWEEN $\mu - 1\sigma$ & $\mu + 1\sigma$: about 68% of data
- BETWEEN $\mu - 2\sigma$ & $\mu + 2\sigma$: about 95% of data
- BETWEEN $\mu - 3\sigma$ & $\mu + 3\sigma$: about 99.7% of data

That's why it is also called 68-95-99.7 rule.

* Where it is APPLIED :-

→ QUICK PROBABILITY ESTIMATES :-

To roughly answer questions like "What percent of students are within 10 marks of the average", when marks are roughly normal.

→ OUTLIER DETECTION :-

Values beyond $\mu \pm 3\sigma$ are very rare ($\sim 0.3\%$), so they are treated as unusual or potential outliers.

→ QUALITY CONTROL & PROCESS MONITORING :-

In manufacturing or process data, if too many points fall outside $\pm 3\sigma$, it signals that the process may be out of control.

→ THE EMPIRICAL RULE WORKS FOR ANY NORMAL DISTRIBUTION, NOT ONLY THE STANDARD NORMAL, STANDARDIZATION JUST MAKES IT EASIER TO SEE AND REMEMBER.

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→ INITIAL DATA CHECK :-

If your data roughly follows 68-95-99.7 around mean, it suggests a normal shape; if not, maybe it is skewed or non-normal.

• The empirical rule helps you build intuition :-

1 σ : Typically middle

2 σ : Still common but getting rare

3 σ : Very Rare/Extreme values.

* EXAMPLE :-

Suppose the distribution of all the plants in a garden is a normal distribution with a mean height of plant being 47.4 inches and standard deviation being 2.4 inches.

Using empirical rule find the total percentage of plants which are 54.6 inches tall.

→ $(100 - 0.15)\% = \underline{99.85\%}$

